

Innovative low-cost approaches to biogas purification in developing economies

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ABSTRACT

The increasing environmental degradation and dependency on fossil fuels have accelerated the global push toward renewable energy. Sustainable Development Goal 7 underscores the critical role of clean, affordable energy in fostering social and industrial development. In developing regions, where organic waste is abundant, biogas—rich in methane—offers a viable renewable energy source. However, raw biogas contains impurities like carbon dioxide and hydrogen sulfide, reducing usability and requiring purification. Conventional purification technologies are often expensive, technologically intensive, and unsuitable for decentralized or small-scale applications in low-resource settings. This review evaluates low-cost, locally sourced materials for biogas purification suited to household and community-level systems. Several affordable materials show promising purification potential: clay removes up to 90 % of contaminants, iron-rich soils up to 93.8 %, steel wool about 95 %, and compost-based media around 80 %.

To further enhance these materials, the study suggests approaches such as drying biomass ash to reduce moisture content, modifying the surface chemistry of activated carbon to improve gas adsorption, combining multiple materials for synergistic effects, and reusing adsorbents to lower costs and environmental impact. Evidence from reviewed literature and pilot projects indicates that these low-cost materials can substantially reduce purification costs and operational burdens. Moreover, some have been successfully implemented in community-level biogas programs, improving energy access in underserved regions. These efforts align directly with the objectives of SDG 7, contributing to cleaner energy transitions and increased energy equity in low-income communities.

Introduction

The global shift toward renewable energy has intensified in response to escalating environmental degradation and the rapid depletion of fossil fuels. Biogas has emerged as a critical solution among sustainable alternatives due to its dual role in clean energy production and organic waste management (Sun et al., 2015). Anaerobic digestion of organic materials, including animal manure, agricultural wastes, and municipal waste, produces biogas, which is mainly made up of methane (CH₄) and impurities, including CO₂, H₂S, and moisture (Kabeyi & Olanrewaju, 2022). In contrast, biogas offers a renewable energy source, particularly

for rural communities lacking grid electricity; its corrosive components—notably CO₂ and H₂S—damage appliances like burners and engines, necessitating purification to enhance safety and efficiency (Alayande et al., 2024). Despite abundant organic waste resources, Biogas potential remains underutilized in sub-Saharan Africa and Southeast Asia.

Countries like Kenya exemplify this gap, where household biogas systems are standard but have minimal purification, limiting applications to basic uses like cooking (Tagne et al., 2021). Small-scale decentralized plants dominate, producing gas quantities too limited for grid integration (Dorich et al., 2022). Equipment corrosion is

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accelerated by unpurified biogas, which has a low energy value. Nevertheless, the utility of centralized large-scale purification could be expanded by enabling grid injection or vehicular fuel use (Mulu et al., 2021a). Current purification technologies face significant barriers, including water scrubbing, pressure swing adsorption, and membrane separation. High operational costs, energy demands, and technical complexity restrict their viability in rural or low-income settings (Niesner et al., 2013). These methods often require skilled labor and infrastructure, making them impractical for household adoption. Consequently, there is an urgent need for affordable, simple systems tailored to small-scale users.

Low-cost purification materials—defined as locally available, economically viable, and environmentally sustainable—present a promising alternative. Derived from organic or agricultural waste (e.g., biochar, zeolites, or lime), these materials require minimal pretreatment and align with circular economy principles. Prior research has focused on advanced techniques like membrane filtration, overlooking cost-effective solutions suitable for decentralized use (Petersson & Wellinger, n.d.). Recent trends, however, indicate growing interest in low-cost alternatives (Fig. 1), underscoring the need for a comprehensive review to evaluate their efficacy, durability, and scalability. With a focus on small-scale systems, this study critically evaluates inexpensive materials for biogas purification. It assesses performance metrics, cost-effectiveness, and reuse potential while addressing variable efficiency and material degradation limitations. Innovations like pretreatment, surface modification, and hybrid material combinations are proposed to enhance functionality. This effort intends to improve sustainable, readily available biogas alternatives for marginalized populations by bridging the gap between academic research and real-world implementation.

Methods

Literature search strategy

A systematic literature search was conducted across five major academic databases: Scopus, Web of Science, PubMed, Google Scholar, and ScienceDirect. The search was designed to capture relevant studies using the following key terms and Boolean combinations: “biogas purification,” “low-cost adsorbents,” “developing countries,” “biogas upgrading,” “household biogas,” “community biogas,” “adsorbent regeneration,” and “cost-effective purification.” The search was limited to studies published in English between January 2000 and December 2023 to ensure both relevance and accessibility.

Inclusion and exclusion criteria

Studies were included to determine if they focused on biogas purification using locally available or low-cost materials and if they reported quantitative performance metrics such as contaminant removal efficiency, regeneration cycles, or cost. Both experimental studies, case studies, and systematic reviews that were contextually relevant to developing economies were considered eligible. Exclusion criteria included studies focusing on large-scale industrial technologies not adaptable to household or community-scale use, patents, conference abstracts, and non-peer-reviewed literature. Articles were also excluded if they lacked sufficient methodological detail or did not report specific performance outcomes.

Screening and selection process

An initial pool of 200 records was retrieved through the database search. Duplicate entries were first removed, and the remaining records underwent title and abstract screening to assess relevance. This was followed by a full-text eligibility review focusing on methodological

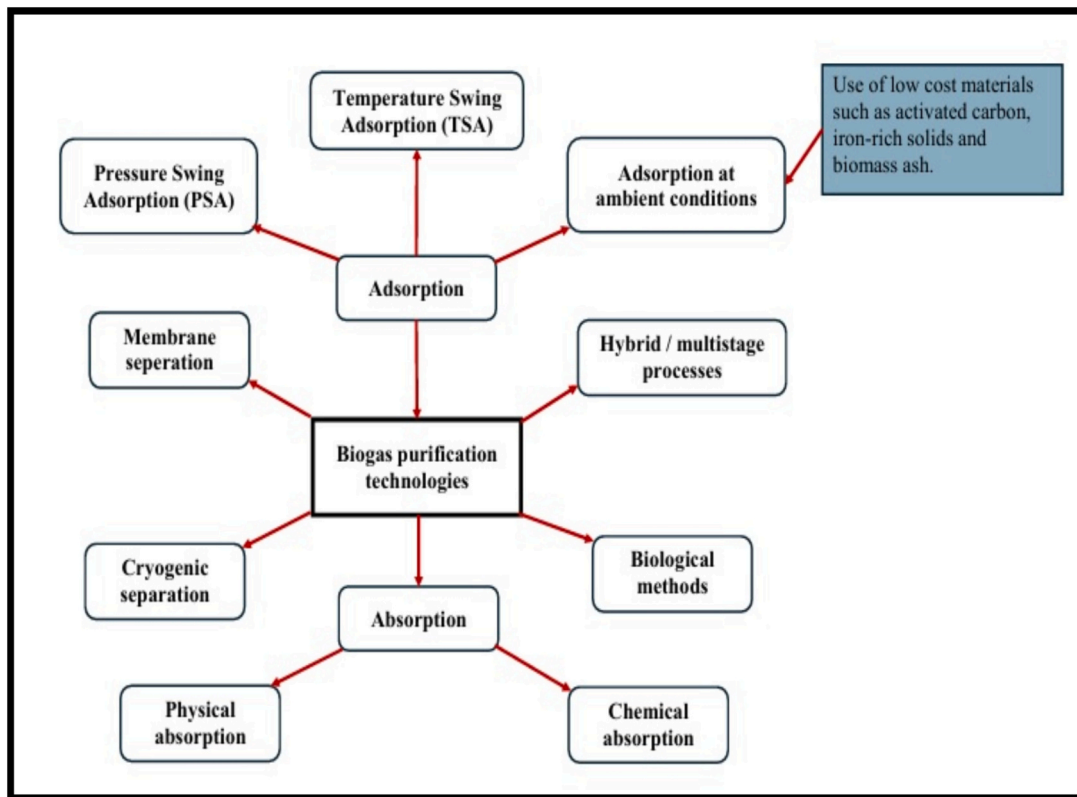


Fig. 1. Illustrates the biogas purification technologies that were examined in this investigation.

quality, data sufficiency, and alignment with the inclusion criteria. Ultimately, 62 articles were included in the final synthesis and review.

Data extraction and quality assessment

Data from the selected studies were systematically extracted and coded based on key variables, including the origin of adsorbent materials, activation methods, operating conditions, removal efficiency, regeneration potential, and economic viability.

A simplified risk-of-bias table adapted from the SANRA (Scale for the Assessment of Narrative Review Articles) framework was applied to evaluate the robustness and credibility of the included studies. This assessment considered clarity in methodological description, reproducibility, and the transparency with which limitations were reported.

The PRISMA 2020 flow diagram (Fig. 2) below summarizes the systematic screening and selection process for studies included in this review. Out of 200 records initially identified through database searches, duplicates were removed, and 143 records underwent title and abstract screening. Following full-text assessment for eligibility and relevance, 62 articles met the inclusion criteria and were included in the final synthesis. This diagram ensures transparency and reproducibility of the review methodology.

Global biogas production

The global push toward renewable energy, driven by climate action and energy transition policies, has spurred significant growth in biogas infrastructure. Derived from anaerobic digestion of organic waste—including agricultural residues (e.g., crop stalks, energy crops), industrial byproducts (e.g., sludge), aquatic biomass (e.g., algae, water hyacinth), and animal manure—biogas offers a versatile energy source (Sher et al., 2024). Its composition and yield vary significantly based on feedstock type, with methane content typically ranging between 50 and 70 % alongside impurities like CO₂ and H₂S (Awe et al., 2017). Recent projections highlight biogas as a cornerstone of future energy systems. By 2040, renewable energy is expected to dominate two-thirds of global power capacity, with biogas, biofuels, and biomass collectively contributing 25 % of renewable energy by 2050 (International Energy Agency, 2021). This growth aligns with the rapid expansion of biogas plants worldwide. Over 32,000 small-to-large-scale biodigesters operate

globally alongside approximately 50 million micro-scale systems used for household cooking and heating (Jameel et al., 2024). Regionally, Europe leads in adoption, accounting for 50 % of global production with a 20 % annual growth rate, followed by Asia (35 %) and the Americas (14 %). Africa, however, lags significantly, highlighting an urgent need for targeted investments and policy support (Bumharter et al., 2023).

In Africa, biogas systems are predominantly small-scale or household-level, addressing basic energy needs such as cooking, lighting, and fertilizer production (Rupf et al., 2016). Households spend roughly 33 h weekly on cooking and 51 h on lighting, relying heavily on traditional biomass like firewood (Rupf et al., 2016). This dependence has detrimental effects on health: indoor air pollution from wood-based stoves causes 2.9 million premature deaths globally yearly, including approximately 450,000 in sub-Saharan Africa. Transitioning to biogas stoves could mitigate these risks by providing cleaner combustion (International Energy Agency, 2021). The International Energy Agency (IEA) emphasizes biogas as a critical solution for clean cooking in developing nations, projecting that by 2040, over 200 million people—half in Africa—will benefit from biogas-driven energy access (International Energy Agency, 2021).

Challenges in biogas purification

The usage of biogas for everyday tasks is growing, particularly in developing countries and rural areas. However, appropriate purification is necessary to optimize its effectiveness and usage. Methane (30–70 % by volume) and carbon dioxide (30–65 % by volume) make up the majority of biogas, with trace amounts of water vapor, hydrogen sulfide (H₂S), hydrocarbons, and nitrogen (Durán et al., 2018; Ali Abd et al., 2024). Removing these impurities increases the methane content, making biogas a more suitable substitute for natural gas in heating, electricity generation, and transportation fuel (Weiland, 2010).

Water vapor and H₂S in unpurified biogas are significant problems since they can cause corrosion in metallic equipment parts (Abatzoglou & Boivin, 2009; Ryckebosch et al., 2011). Furthermore, serious health hazards are associated with H₂S exposure, such as eye irritation, exhaustion, appetite loss, and respiratory problems.

Exposure to H₂S can have serious health consequences, such as eye damage, sudden collapse, and even death, at concentrations between 150 and 2000 ppm. According to regulatory safety guidelines, exposure

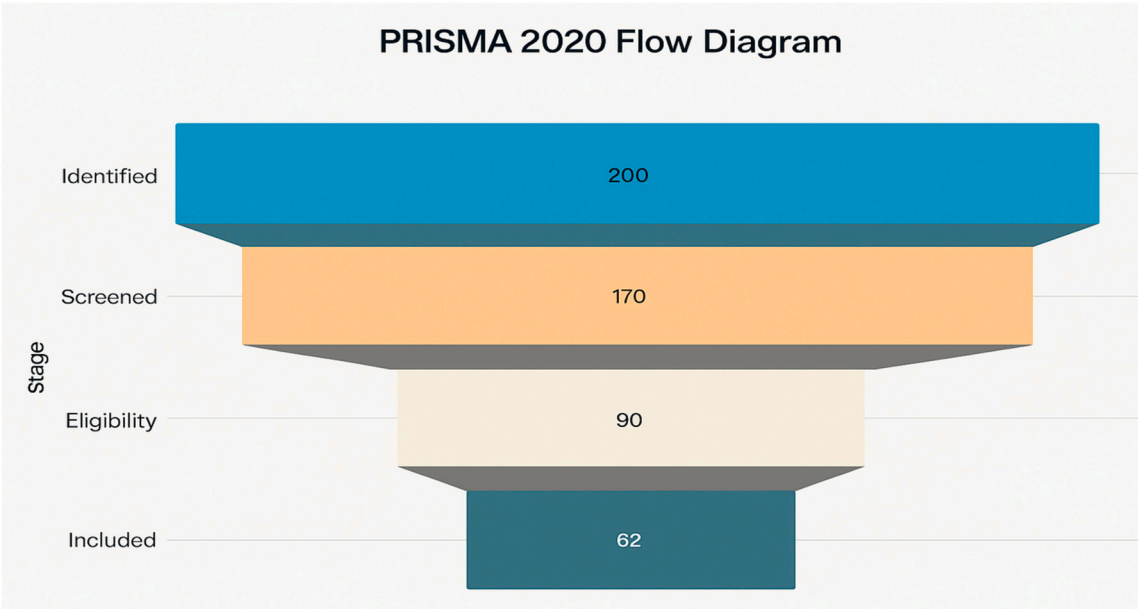


Fig. 2. PRISMA 2020 flow diagram illustrating the literature selection process for the review of low-cost biogas purification methods.

should be restricted to 10–50 ppm, depending on the environment (Chan et al., 2022). CO₂ also impacts biogas quality by reducing its calorific value and causing freezing in valves and metering points (Wylock & Budzianowski, 2017). These challenges highlight the necessity of purifying biogas before its use. Numerous purification technologies exist, including adsorption, membrane separation, scrubbing, absorption, and biofiltration (Upadhyay et al., 2023). Among these, adsorption can be the most cost-effective for small-throughput rural digesters (<100 m³ day⁻¹), due to its simplicity, low energy demand, and compatibility with locally available materials (Abdullah et al., 2017). Biogas purification adsorbent materials are widely available and affordable (Skerman et al., 2017). Many of these materials are naturally sourced and can be obtained locally, making them suitable for low-cost purification systems (Pham et al., 2019). Iron-rich soils, activated carbon, biomass ash, and kaolin are some of the most promising adsorbents; they have all demonstrated efficacy in extracting CO₂ and H₂S from biogas (Papurello et al., 2019; Barelli et al., 2017).

Purification expenses are significantly decreased by using natural materials like zeolites (Fontseré Obis et al., 2017). Similarly, because of its large surface area, potent adsorption ability, and affordability, activated carbon from organic waste is a desirable choice (Mrosso et al., 2020). Other low-cost candidates for desulfurization include iron-rich soils and kaolinite, both readily available and inexpensive (Abdullah et al., 2017). Despite the potential of these materials, research on their application in biogas purification remains limited (Mrosso et al., 2024b).

Furthermore, the efficiency of inexpensive materials for biogas purification in impoverished nations has not been thoroughly assessed by any study. By examining methods to improve the performance of inexpensive purification materials and evaluating their application, this review seeks to close that gap. By identifying cost-effective and efficient solutions, this study makes biogas a more viable energy source, particularly for low-income communities.

Existing technologies for purifying biogas and their drawbacks

The most popular techniques for purifying biogas are membrane separation, adsorption, and absorption; however, other technologies have also been developed and brought to market (Aminuddin et al., 2024). The necessary purification efficiency, operational circumstances, and related expenses are some variables that influence the choice of an appropriate purification technique (Ali Abd et al., 2024). However, comparing these technologies can be challenging since specific process conditions and the intended use of the purified biogas influence their performance. This section examines the existing biogas purification techniques, highlighting their advantages and limitations.

Absorption technologies

Absorption-based biogas purification involves the removal of impurities through either physical or chemical absorption.

Technologies of physical absorption

Contaminants in biogas, such as CO₂, H₂S, and NH₃, are dissolved in a cleansing liquid during physical absorption due to their superior solubility in methane (CH₄). Water and organic solvents are the most frequently employed cleansing liquids. This approach can handle various pollutant concentrations and accomplish high removal efficiencies, often reaching 98.5 %. Nevertheless, it necessitates additional treatment of the discarded absorbent, consumes substantial quantities of water and/or chemicals, and necessitates a significant initial investment (Angelidaki et al., 2018).

Water scrubbing

Since CO₂ and H₂S are more soluble in water than methane, they are frequently removed during water purification. About 41 % of all biogas

purification systems worldwide employ this popular industrial-scale refining technique. A separation technique that is particularly effective at lower temperatures is the solubility of CO₂ in water, which is approximately 26 times greater than that of CH₄ at 25 °C (Awe et al., 2017). Once absorbed, the contaminants can be desorbed by reducing the pressure to 1 bar, allowing for water regeneration. This method is praised for its simplicity, reliability, and high methane recovery rates (often exceeding 97 %) (Herdem et al., 2024).

Despite its effectiveness, water scrubbing has several drawbacks, including high initial investment costs, sensitivity to variations in gas input, and significant water and energy consumption. Additionally, it requires gas drying before final use. The need for high pressure and low temperatures increases operational costs, making this method less accessible for low-income users. Furthermore, bacterial growth on the packing material can cause clogging, further complicating maintenance and operation (Ghaib & Ben-Fares, 2018).

Organic solvent scrubbing

This method uses organic solvents, such as methanol, polyethylene glycol, and dimethyl ethers of polyethylene glycol, in place of water to absorb impurities. These solvents have a higher capacity for dissolving CO₂ than water, reducing the required circulation rate and equipment size for processing the same raw biogas (Angelidaki et al., 2018). In addition to CO₂, organic solvents can absorb H₂S, NH₃, and moisture. While organic solvent scrubbing is more efficient than water scrubbing, it has significant limitations. The main challenges include high investment and operational costs and the difficulty of regenerating the used solvent. Complete solvent regeneration typically requires boiling, which adds to the process's energy demands and overall expense (Awe et al., 2017). Although this method offers improved performance, its costly handling and solvent regeneration requirements make it less feasible for small-scale and low-income applications (IEA, 2018).

Technologies of chemical absorption

One method of purifying biogas is chemical absorption, which uses chemical reactions between the absorbent and the desired contaminants. The most commonly used chemical absorbents are amine-based solutions, such as methyl diethanolamine and dimethyl ethanolamine, and alkali solutions, such as potassium hydroxide, sodium hydroxide, potassium carbonate, iron hydroxide, and iron chloride (Lasocki et al., 2015).

Amine scrubbing

The method of amine scouring is frequently employed to capture CO₂ and H₂S from biogas using aqueous solutions of alkyl amines. This process facilitates the regeneration of the amine solution by boiling and removing the adsorbed gases. One of its significant advantages is its ability to achieve high methane enrichment, often exceeding 99 %, while maintaining relatively low operational costs. Research on amine-based biogas upgrading using diglycolamine, methyl diethanolamine, and diethanolamine has demonstrated up to 99 % CO₂ removal efficiency, yielding a biomethane product with approximately 91 % methane content (Cavaignac et al., 2021). Despite its efficiency, amine scrubbing has several limitations. The high cost of amine solvents and evaporation-induced solvent loss raises operating costs. Amines also pose toxicity hazards to the environment and people. Other issues include salt precipitation, foaming, corrosion of equipment, and amine degradation due to oxygen exposure. The regeneration of absorbents requires significant energy input, making this process less attractive for cost-sensitive applications (Angelidaki et al., 2018).

Inorganic solvent-based absorption

Aqueous ammonia solutions, sodium hydroxide (NaOH), and sodium

carbonate (Na_2CO_3) are examples of inorganic solvents that effectively absorb moisture, CO_2 , and H_2S and can be used to purify biogas. Iron-based solvents that specifically target the elimination of H_2S include iron hydroxide ($\text{Fe}(\text{OH})_3$) and iron chloride (FeCl_2) (Meena & Awale, 2024). However, sulfur produces irreversible salts when combined with inorganic solvents, such as $\text{Ca}(\text{OH})_2$ and NaOH . Similarly, iron chloride generates insoluble iron sulfide, which must be disposed of and cannot be reused. Conversely, iron hydroxide and ferric chelate make partial regeneration possible through thermal treatment or air oxidation (Miltner et al., 2017). Aqueous alkaline salts are frequently chosen because they are more widely available and less expensive than amines.

For moisture removal, glycol and hygroscopic salts have been utilized, but their high costs make them impractical for large-scale use (Meena & Awale, 2024). Research has shown that NaOH and $\text{Ca}(\text{OH})_2$ achieve H_2S removal efficiencies of 88.61 % and 89.87 %, respectively, slightly lower than the 92.41 % efficiency observed with steel wool (Pallavicini et al., 2023). These substances are beneficial when an increase in the heating value of biogas is sought, even if they also efficiently remove CO_2 (Kulkarni & Ghanegaonkar, 2019). Under certain circumstances, silica gel, a commonly used desiccant, can absorb moisture and sulfur compounds, enhancing the quality of biogas.

Advantages and limitations

Chemical absorption offers high biomethane recovery rates (greater than 97 %) and efficient removal of CO_2 and H_2S . However, its drawbacks include high energy demands for regeneration, salt precipitation, foaming, and oxygen poisoning of amines, which can cause column clogging (Bauer et al., 2013; Deng & Hägg, 2010). Additionally, the environmental impact of waste byproducts and the corrosive nature of these chemicals reduce the sustainability of this purifying method (Mathias, 2014).

Membrane separation

One method for separating the components of biogas is membrane separation, which depends on the membranes' selective permeability. This technology is frequently utilized for biogas purification because of its lower energy usage, simplicity of operation, and installation convenience. In addition to having a high methane recovery rate of over 96 % and an efficiency of up to 98 % in eliminating hydrogen sulfide (H_2S), membrane modules are lightweight, which lowers expenses (Angelidaki et al., 2018). Most membrane materials are designed to allow the passage of carbon dioxide (CO_2), water, and ammonia while retaining methane (CH_4) (IEA, 2018). With CO_2/CH_4 selectivity values of up to 87 %, advances in membrane technology, including mixed matrix membranes and 6FDA-based polyimides, have significantly increased CO_2 permeability, which may now reach up to 18,500 barrer (Rodero et al., 2024).

Membrane separation has drawbacks despite its benefits. It takes several phases of separation to get high-purity biomethane, which adds complexity. Because they are delicate, membranes must be handled carefully to avoid damage (Zhao et al., n.d.). Membranes offer high-purity methane but are expensive to buy, maintain, and run slowly (Meena & Pal, 2024). Commercial membrane-based systems are less practical for low-income areas because of these issues.

Cryogenic separation

The purification process of cryogenic separation uses variations in the boiling points of the constituents of biogas. Impurities such as CO_2 , H_2S , nitrogen (N_2), oxygen (O_2), and siloxanes condense or freeze when raw biogas is subjected to temperatures as low as -155°C and more than 85 bar pressures. Meanwhile, methane stays in the gas phase. After that, these condensed contaminants can be separated into liquid and solid portions (Awe et al., 2017). The capacity to generate high-purity

biomethane with concentrations between 90 % and 99 % is a key benefit of cryogenic separation. This purification procedure is environmentally beneficial since no chemical reagents are needed (Zhang et al., 2017).

However, because cryogenic separation requires sophisticated cooling and compression equipment, it is very energy-intensive and requires a significant initial investment and continuous maintenance expenditures. The complexity of the equipment and its high operational costs make it impractical for small-scale or low-income applications (Zhang et al., 2017). In a study focused on removing siloxanes from biogas using cryogenic condensation, researchers achieved a remarkable 99.87 % removal efficiency, demonstrating the effectiveness of this method for specific purification needs (Piechota, 2021).

Condensation

A physical process called condensation is used to extract moisture from unprocessed biogas. The process involves compressing the gas before cooling it down and then expanding it to a target temperature. As the gas cools, water droplets condense and separate from the biogas (Ryckebosch et al., 2011).

Even though this procedure is straightforward, it requires high working pressures—typically 6 to 11 bars—to achieve a lower dew point of roughly -40°C , making removing moisture easier.

Due to these high-pressure requirements, condensation is both energy-intensive and costly.

There are three primary methods of physical moisture separation:

1. Cyclone separators — These use centrifugal force to remove condensed water droplets.
2. Water traps — These are specially designed pipe systems that collect and drain water droplets.
3. Moisture traps utilize pressure differences to generate a low-temperature environment that promotes water condensation (Atelge et al., 2021).

Biological methods

Biological purification techniques utilize specialized microorganisms to remove impurities from biogas. These bacteria, which transform hydrogen sulfide (H_2S) into elemental sulfur and sulfates, include *Halothiobacillus*, *Thiobacillus*, *Paracoccus*, and *Acidithiobacillus*. This process is known as biological desulfurization. In addition to H_2S removal, these bacteria consume carbon dioxide (CO_2) as a carbon source, contributing to biogas purification (Awe et al., 2017). However, this process requires careful monitoring of reactor pH levels, as fluctuations can negatively impact methane production. Additionally, sulfur-rich byproducts must be removed regularly to prevent accumulation (Ryckebosch et al., 2011).

Another biological approach involves *Chlorobium limicola*, an anaerobic phototrophic bacterium that oxidizes H_2S in light and CO_2 . However, a disadvantage of biological techniques is the rise in nitrogen content brought on by air dosing, which restricts the use of such purified biogas in engine and vehicle applications because of residual oxygen (IEA, 2018). Biofiltration is another effective biological purification method. Using particular bacteria to oxidize H_2S can recover up to 96 % of the methane, remove 96 % of the CO_2 , and ultimately eliminate H_2S (Ryckebosch et al., 2011). This method is comparatively easy to use and requires additional tools, including pressurized tanks (IEA, 2018). High efficiency has been shown in studies on anoxic desulfurization in biotrickling filters, with a removal rate of 99 % for H_2S and an elimination capacity of $287\text{ gS/m}^3/\text{h}$ (Almenglo et al., 2023).

Additionally, biological methane enrichment is a promising approach for enhancing methane content in biogas. This process leverages *hydrogenotrophic methanogens*, which utilize CO_2 as a carbon source and electron acceptor while using hydrogen (H_2) as an electron donor.

This microbial conversion effectively increases methane yield while reducing CO₂ levels in biogas (Barelli et al., 2017).

Techniques for adsorption

One of the most sensible and cheap methods for cleaning biogas is adsorption (Abdullah et al., 2017). This technique relies on the ability of adsorbent materials to capture specific gas molecules based on their molecular size selectively. Since CO₂, H₂S, and H₂O have smaller molecular sizes compared to CH₄, they tend to adhere more readily to the surface of adsorbents, facilitating their separation from methane. Many variables affect adsorption efficiency, including temperature, pressure, and the adsorbent material. Zeolites, zinc oxide, aluminum oxide, silicates, silica gel, and iron oxides are common adsorbents used in biogas purification (Miltner et al., 2017).

Pressure swing adsorption (PSA), temperature swing adsorption, and vacuum swing adsorption are only a few adsorption techniques. Because PSA uses less energy and resources than other adsorption techniques, it is frequently used to separate CO₂ and H₂S (IEA, 2018). PSA is based on the principle that gases adhere to solid surfaces under pressure, whereas adsorbed gases escape when pressure is lowered. This allows for selective removal of CO₂, N₂, O₂, and H₂S from biogas (Angelidaki et al., 2018). The most effective adsorbents for eliminating CO₂ from PSA systems are metal-organic frameworks, zeolites, and molecular sieves (Durán et al., 2018). However, H₂S tends to bind irreversibly to molecular sieves, reducing their reusability. To maintain the efficiency of PSA, it is recommended to remove H₂S before CO₂ adsorption (IEA, 2018).

The lifespan of the adsorbent material is significantly reduced if CO₂ and H₂S are absorbed simultaneously (Miltner et al., 2017). To avoid adsorbent contamination, water vapor must be eliminated before biogas purification (Ardolino et al., 2021). Despite its advantages, PSA is more complex than straightforward methods like water scrubbing. It involves multiple stages to achieve CH₄ separation under pressure (Noorain et al., 2019). PSA's main advantages are the capacity to supply gas at high pressures, excellent energy efficiency, and high bio-methane recovery rates (94–98 %). The main disadvantages are the requirement for exact temperature and pressure control, expensive startup and operating expenses, and limited viability for small-scale applications (Noorain et al., 2019).

At room temperature, adsorbents such as silica gel and activated carbon efficiently absorb siloxanes and moisture. Magnesium oxide or aluminum oxide can also be used to dehydrate biogas. Adsorbents such as iron (III) oxide, steel wool, zeolite, and activated carbon have been effectively removed (Ryckebosch et al., 2011). Studies on fixed-bed adsorption using commercial steel wool have demonstrated 95 % H₂S removal efficiency (Kulkarni & Ghanegaonkar, 2019). Particularly successful is iron oxide-based adsorption, which may remove up to 99.98 % of the material and lower H₂S levels to less than 1 ppm. Inappropriate heat dissipation during regeneration might result in an ignited reaction because of the high exothermic reaction and running costs (Ryckebosch et al., 2011). Research indicates that 1 kg of iron oxide can remove between 0.2 and 0.716 kg of H₂S (Xie et al., 2023).

Additionally, activated carbon, molecular sieves, and membranes can effectively adsorb oxygen and nitrogen from biogas (IEA, 2018). For H₂S removal, 1 kg of activated carbon can adsorb between 0.2 and 0.5 kg of H₂S (Egger et al., 2023). To make adsorption more accessible and cost-effective for small-scale applications, especially in developing countries, further exploration into multistage adsorption processes and locally available materials is necessary to reduce operational costs and improve efficiency.

Multistage, hybrid, and combined processes

Combining many purification methods in a multistage system can increase biogas purification efficiency. These hybrid strategies combine

readily available raw materials with straightforward designs, which allows them to be modified for local biogas upgrading. The removal effectiveness of H₂S, for instance, was 96.84 % in a packed column containing activated carbon granules, monoethanolamide, and calcium hydroxide.

However, the toxicity of some of the chemicals utilized and the significant energy consumption for chemical regeneration are the main disadvantages of this approach (Kulkarni & Ghanegaonkar, 2019). In another study, the concentration of H₂S was successfully reduced from 0.50 % to 0.02 %, CO₂ from 35 % to 20 %, and moisture from 0.90 % to 0 % using a packed bed system that used activated charcoal to remove H₂S and local potash to remove CO₂ and humidity. Despite these improvements, the system's CO₂ removal efficiency was inadequate [55]. A mixed absorption and adsorption system comprising CaO, FeO₃, FeCl₂, activated carbon, silica gel, and NaSO₄ significantly reduced the CO₂ content to less than 7 %. Depending on whether FeCl₂, Fe₂O₃, or activated carbon was employed, H₂S concentration dropped by 85–95 %. Additionally, silica gel and Na₂SO₄ effectively lowered water vapor content to 0.2 % (Al Mamun & Torii, 2017).

Another hybrid method that used dry lime, potassium hydroxide, and iron oxide (from lathe machine chips) had an H₂S removal effectiveness of 85.5 % and a CO₂ removal efficiency of 90 % (Álvarez-Gutiérrez et al., 2016). Combining oxidized steel wool for H₂S removal, water for CO₂ absorption, and silica gel for moisture removal, a different method raised the methane concentration from 65 % in raw biogas to 92 % in bio-methane [58]. In another study, iron oxide and iron chloride were combined to remove 88 % of the CO₂ and 98 % of the H₂S (Al Mamun & Torii, 2017). A cost-effective biogas filtration system was created, integrating absorption and adsorption processes utilizing water, NaOH, and silica gel as absorbents, with iron sponge and activated carbon as adsorbents (Mojica et al., 2018). The system successfully eliminated approximately 72 % of non-combustible gases, leading to a 54.38 % increase in bio-methane content. This study demonstrated the potential of using cost-effective and readily available materials for efficient biogas purification.

A potential solution: low-cost adsorption materials

Affordable, easily accessible, and efficient adsorbents used to upgrade biogas effectively are referred to as low-cost materials for biogas purification (Skerman et al., 2017). These materials are typically derived from locally sourced and naturally occurring substances, making them a cost-effective alternative for biogas treatment, especially in low-income regions (Pham et al., 2019). Waste iron fillings, calcined eggshells, biomass ash, and activated carbon are among the materials that can be used to purify biogas (Pham et al., 2019; Meena & Amit Pal, 2022a).

These materials help lower purification costs and contribute to eco-friendly waste management by repurposing substances that would otherwise be discarded (Finnveden et al., 2007). This section examines how these inexpensive ingredients are prepared and used for biogas purification. Fig. 3 presents various low-cost absorbent materials used for biogas purification. These include waste iron chips from workshops, red soil, biomass ash, and activated carbon derived from organic waste. These materials were selected for their affordability and potential effectiveness in removing impurities from biogas.

Activated carbon

The enormous surface area, microporosity, and potent adsorption activity of activated carbon make it a popular adsorbent in biogas purification. Organic materials such as coconut shells, nutshells, sunflower seeds, maize cobs, straw, pine sawdust, and forest leftovers are the primary sources of carbonization (Meena & Amit Pal, 2022a). These abundant and inexpensive raw materials make activated carbon a practical choice for biogas purification. By improving its adsorption

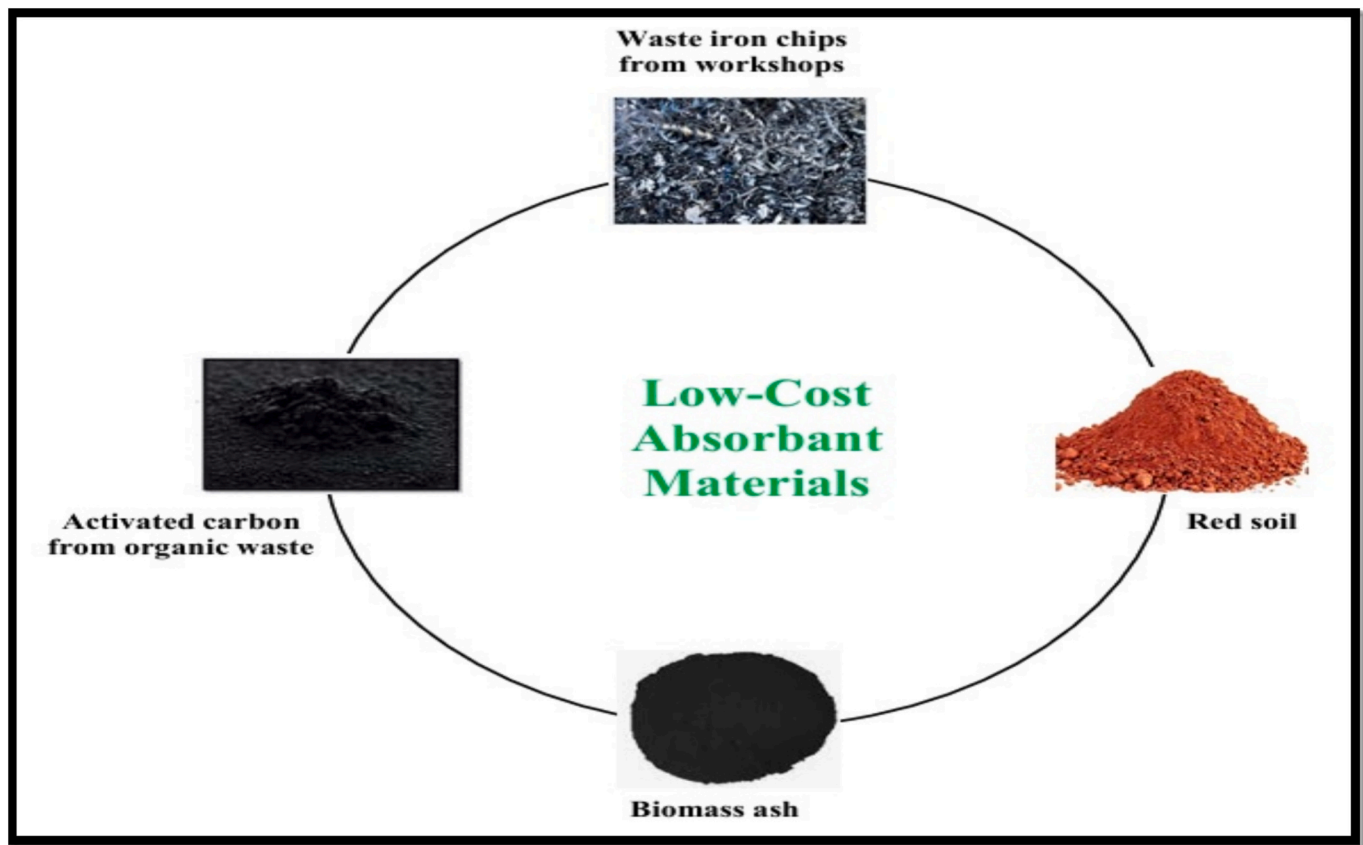


Fig. 3. Illustrates the low-cost biogas purification materials examined in this study.

capabilities, the activation process efficiently extracts CO_2 and H_2S from biogas (Wu et al., 2015).

Depending on the material source and processing parameters, physical or chemical activation can increase the porosity of activated carbon. Optimized activation procedures can attain a pore capacity of $90 \text{ cm}^3/\text{g}$ and a surface area of up to $1400 \text{ m}^2/\text{g}$ (Mrosso et al., 2020). The raw materials must be cleaned, dried, and carbonized in an inert environment at temperatures lower than 800°C . The material is then activated using either physical methods (such as steam or gas activation) or chemical methods (using reagents like potassium iodide or zinc acetate) (Mdoo, n.d.). Impregnating activated carbon with specific chemicals or metal oxides further enhances its adsorption efficiency by increasing the number of active sites available for gas capture (Durán et al., 2018). This characteristic renders activated carbon extremely efficient for upgrading biogas, providing better H_2S removal capabilities than zeolites and ionic sieves (Micoli et al., 2014; Sisani et al., 2014). Additionally, activated carbon can be regenerated and reused, making it a sustainable and cost-effective option for biogas purification. However, selecting the right source material is crucial to minimize secondary waste generation and ensure efficient purification (Meena et al., 2023).

Soils abundant in iron

Iron-rich red soils have demonstrated considerable promise as adsorbents for biogas desulfurization, especially when eliminating hydrogen sulfide (H_2S) from biogas streams (Pham et al., 2019). Their cost-effectiveness, ease of use, and sustainability make them a viable alternative for biogas purification, especially in low-income regions (Skerman et al., 2017). The key to their effectiveness lies in the presence of iron oxides, which react with H_2S in biogas to form iron sulfide (Skerman et al., 2017). This reaction helps remove H_2S efficiently, and the adsorbent can be regenerated by simple exposure to air, restoring its

ability to remove H_2S . The soil's porosity and texture are essential factors in adsorption effectiveness. Compared to sandy or loamy soils, soils with a higher clay content have a larger surface area, which improves their adsorption capability.

Research indicates that clay-based materials have an adsorption capacity of 0.051 mg/g for H_2S and 232.1 mg/g for CO_2 , which allows them to remove 100 % of H_2S from biogas. Because CO_2 and H_2S have smaller molecular sizes than methane and can more easily enter adsorbent pores, our results imply that clay can remove these gases from biogas (Mulu et al., 2021a). Magnetite ($\text{FeO}(\text{OH})$), goethite ($\text{FeO}(\text{OH})$), and haematite (FeO_3) are found in iron-rich red soils and help remove H_2S (Mrosso et al., 2020). Research on the biogas purification potential of iron-rich soils in New Zealand showed that allophanic soil had a 100 % H_2S removal efficiency, brown soil had a 92.5 % removal efficiency, and black sand had a 96.2 % removal efficiency. Additionally, during regeneration, brown soil demonstrated increased efficiency (99.10 %), demonstrating the iron-rich soils' reusability in biogas upgrading (Meena & Pal, 2024). The efficiency of H_2S removal often increases with repeated regeneration cycles, making these soils a sustainable solution for biogas purification (Meena & Amit Pal, 2022b).

Alternative sources of iron oxide

Iron oxides can be found in various natural and anthropogenic sources, including acid mine drainage and naturally existing iron-rich soils (Wei & Viadero, 2007; Andersen et al., 2012). These alternative sources provide an additional supply of iron oxides for biogas desulfurization. One of the most effective low-cost adsorbents is oxidized steel wool, which transforms into iron oxide and achieves 95 % H_2S removal efficiency when exposed to atmospheric conditions. This material is easily regenerated, enhancing cost-effectiveness (Ryckebosch et al., 2011). A similar level of performance was observed in fixed-bed

systems, where oxidized steel wool successfully maintained 95 % removal efficiency even after multiple regeneration cycles (Kulkarni & Ghanegaonkar, 2019).

Another promising source of iron oxides is iron chips from lathe machines, which are often discarded as industrial waste. After being exposed to air, these chips can purify biogas. Raw biogas reacts with iron chips or oxidized steel wool to produce elemental sulfur when the iron oxides react with H₂S (Egger et al., 2023). This process removes harmful sulfur compounds and promotes waste recycling, making it a sustainable and low-cost alternative for biogas upgrading.

Biomass ash

A popular low-cost adsorbent for purifying biogas is biomass ash, which is made from various organic and waste sources. It can be divided into three types: coal ash from burning coal, biomass ash from plant wastes like wood, and waste ash from municipal solid waste. Since ash is typically a waste byproduct, its use in biogas purification offers a low-cost and environmentally friendly alternative (Chavez & Guadarrama, 2015). Limited trials suggest its potential to remove not only CO₂ and H₂S—where removal efficiencies of up to 85 % have been reported—but also trace contaminants like siloxanes, sulfur compounds, and chlorinated volatile organic compounds. However, further studies are needed to quantify these effects.

Effective H₂S removal was shown in studies on landfill gas desulfurization utilizing bottom ash from municipal solid waste, and the wasted adsorbent could be stabilized for safe disposal or used as a source of nutrients for farming (Fernández-Delgado Juárez et al., 2018). Landfill biogas has also been purified using biomass ash from municipal trash incinerator bottoms, with significant H₂S adsorption capabilities of 75–120 g H₂S/kg of dry bottom ash (Ducom et al., 2009; Turgeon et al., 2017). According to a different study, biomass ash is an efficient and reasonably priced material for desulfurization, with the ability to adsorb 0.50 to 1.30 kg H₂S/tonne of ash (Fernández-Delgado Juárez et al., 2018).

In addition to ash, cow manure compost has been explored for its potential in H₂S removal. Experiments showed that it could achieve an 80 % removal efficiency for biogas containing 1500 ppm H₂S (Laabidi et al., 2023). These findings indicate that biomass-derived materials can be valuable for biogas upgrading, particularly in regions with limited access to expensive purification technologies.

Assessment of the efficacy of economical materials

Hydrogen sulfide (H₂S) removal

Removing hydrogen sulfide (H₂S) from biogas is essential to prevent equipment corrosion and enhance the gas's quality and usability. Using low-cost adsorbents for H₂S removal is affordable and results in high-purity biomethane. Factors such as the biogas flow rate, adsorbent mass, and residence period affect how well biogas desulfurization works. The usefulness of inexpensive materials in biogas purification is demonstrated by studies that exhibit removal efficiencies ranging from 85 % to 99 %. One advantage of low-cost adsorbents is their reusability, allowing various applications. After use, biomass ash can be recycled in agriculture (Fernández-Delgado Juárez et al., 2018). Iron-rich soils and iron filings can be revitalized by merely exposing them to air, and their capacity to absorb materials for later use is restored (Mrosso et al., 2020). However, spent activated carbon presents challenges in regeneration, making it less sustainable than other low-cost alternatives. The effectiveness of various low-cost materials in biogas desulfurization is presented in Table 1 and Fig. 4.

Removal of carbon dioxide (CO₂)

Biogas's energy content is decreased by carbon dioxide (CO₂), so

Table 1

Effectiveness of low-cost adsorbents for H₂S removal from biogas.

Substance	Execution	Reference
Bottom ash from municipal solid waste incineration (MSWI)	Adsorbs 2.8 g H ₂ S/kg of ash	(Ducom et al., 2009)
MSWI bottom ash	Adsorption capacity of 120 ± 25 g H ₂ S/kg	(Turgeon et al., 2017)
Ash from biomass	Can capture 1.30 kg H ₂ S/ton of ash	(Fernández-Delgado Juárez et al., 2018)
Activated carbon with potash	Reduces H ₂ S concentration from 0.50 % to 0.01–0.03 %	(Tian et al., 2020a)
Activated carbon	Adsorbs 6.9 mg H ₂ S/g of carbon	(Mrosso et al., 2020)
Activated carbon	Lowers H ₂ S from 0.50 % to 0.02 %	(Orhorhoro et al., 2018)
Red soil	A capacity of 2.12 g H ₂ S/kg for adsorption	(Skerman et al., 2017)
Iron-rich soil	Removes 90–99 % of H ₂ S	(Pham et al., 2019)
Red rock	Achieves 95 % H ₂ S removal	(Mrosso et al., 2020)
Iron fillings	Removes 90 % of H ₂ S	(Tian et al., 2020b)
Cow manure compost	Achieves 80 % removal efficiency	(Laabidi et al., 2023)
Household waste slag	Removes more than 98 % of H ₂ S	(Petersson & Wellinger, n.d.)
Activated carbon (unmodified)	H ₂ S adsorption capability of 21 g/kg	(Abatzoglou & Boivin, 2009)
Impregnated activated carbon	Adsorption capacity of 150 g H ₂ S/kg	(Abatzoglou & Boivin, 2009)
MSWI bottom ash	Removes 99–99.5 % of H ₂ S	(Mostbauer et al., 2014)
Nanostructured iron oxide	Adsorbs 2.5 mg H ₂ S/g	(Cristiano et al., 2020)
Sweet potato leaves are the source of activated carbon.	The capacity of 3.5 g sulfur to adsorb 100 g of carbon	(Juma et al., 2020)
Soda ash	Achieves 94 % removal efficiency	(Mrosso et al., 2023)

removing it is essential to raising the gas's overall quality and heating value. Activated carbon is widely used among low-cost adsorbents due to its high adsorption capacity. Studies have shown that activated carbon can adsorb up to 5.24 mol CO₂/g, highlighting its effectiveness in CO₂ removal (Durán et al., 2018). The efficiency of CO₂ removal depends on three key factors: biogas flow rate, adsorbent mass, and residence time. A slower biogas flow rate enhances gas-adsorbent interaction, increasing adsorption efficiency. Additionally, a larger quantity of adsorbent provides more active sites for CO₂ capture, while a longer residence time ensures sufficient contact for effective adsorption (Fernández-Delgado Juárez et al., 2018). Table 2 summarizes various low-cost materials used for CO₂ removal and their respective performance levels.

Improving the efficacy of economical materials: prospective outlook

Improving their durability and efficiency is crucial to enhancing the industrial adoption of inexpensive materials for biogas purification. This aligns with Sustainable Development Goal 7 (SDG 7), which promotes affordable and clean energy. The performance of these materials can be optimized through five key strategies: refining preparation conditions, implementing effective pre-treatment methods, modifying surface properties, combining multiple materials, and regenerating and reusing spent materials. These approaches collectively enhance the effectiveness and sustainability of biogas purification processes.

Optimizing material preparation conditions

The efficiency of low-cost adsorbents is highly dependent on their preparation methods. In the case of biomass ash, the ash fusion

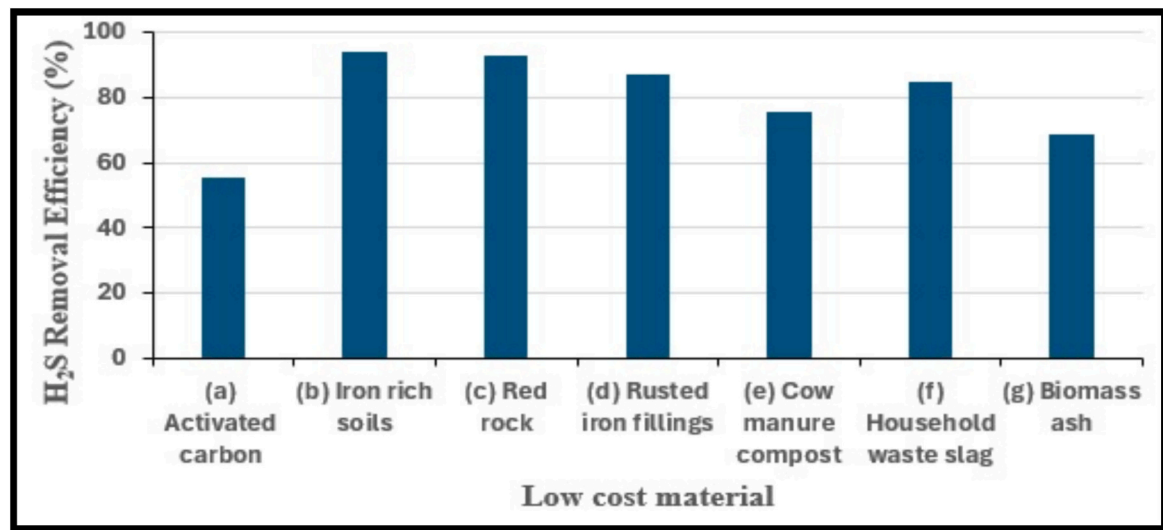


Fig. 4. Study comparing biogas desulphurization using different inexpensive adsorbents (a) (Orhorhoro et al., 2018), (b) (Pham et al., 2019), (c) (Mrosso et al., 2020), (d) (Meena & Amit Pal, 2022b), (e) (Laabidi et al., 2023), (f) (Petersson & Wellinger, n.d.), and (g) (Fernández-Delgado Juárez et al., 2018).

Table 2
Effectiveness of low-cost materials for CO₂ removal from biogas.

Material	Performance	Reference
Activated carbon	Adsorbs 5.24 mol CO ₂ /g of carbon	(Durán et al., 2018)
Activated carbon	Increases methane concentration from 62 % to 91 %	(Al Mamun et al., 2016)
Activated carbon with potash	Lowers CO ₂ levels from 30.00 % to 17.90 %	(Kriauciūnas et al., 2021)
Biomass ash (wood ash)	Captures 35–135 kg CO ₂ /ton of ash	(Fernández-Delgado Juárez et al., 2018)
MSWI bottom ash	Adsorbs 11 g CO ₂ /kg of ash	(Ducom et al., 2009)
MSWI bottom ash	Captures between 10.5 and 22.9 kg CO ₂ /ton of ash	(Mostbauer et al., 2014)
Activated carbon (from coconut shell)	Adsorbs 48 cm ³ CO ₂ /g of carbon	(Sarkar & Bose, 1997)
Activated carbon	Reduces CO ₂ levels by 45 %	(Lasocki et al., 2015)
Calcined eggshells	Achieves 82.5 % reduction in CO ₂	(Mrosso et al., 2024a)
Wood ash	Reduces CO ₂ concentration by 67.9 %	(Mulu et al., 2021b)

temperature plays a crucial role in adsorption performance. Studies indicate that at 550 °C, biomass ash retains unburned carbon, resulting in poor adsorption efficiency, whereas heating above 800 °C removes all carbon, producing a more effective adsorbent for H₂S removal (Skerman et al., 2017). Similarly, size reduction is essential for improving the adsorption capacity of iron-rich red soil. Crushing and sieving the soil to particles smaller than 2 mm significantly increases its surface area, enhancing its H₂S adsorption capability (Skerman et al., 2017).

Strategies for pretreatment

Materials can be pre-treated to improve adsorption efficiency and eliminate contaminants before usage. One technique is to air-dry biomass ash, which enhances its adsorbent properties by lowering its moisture content and preventing undesired phase development (Ducom et al., 2009). Adjusting its moisture content to around 15–25 % for bottom ash enhances H₂S removal efficiency. Since H₂S is hydrophilic, some moisture allows for better dissociation of the gas, leading to improved adsorption through physicochemical interactions (Fernández-Delgado Juárez et al., 2018). In the case of red soil, pre-treatment by

removing plant debris and drying at room temperature is necessary to enhance its adsorption properties.

Surface modification

Altering the surface properties of adsorbents can significantly increase their adsorption capacity. For instance, treating biomass ash with a CO₂/N₂ gas mixture (50 % CO₂/50 % N₂) until saturation leads to hardening and improved maturation, which enhances H₂S removal efficiency (Turgeon et al., 2017). For activated carbon, surface area and porosity can be increased through size reduction and activation. Activation can be done physically (e.g., heating at high temperatures) or chemically (e.g., using acid or alkaline treatments) to increase the number of adsorption sites (Vilella et al., 2017). Modifying activated carbon by introducing nitrogen-containing functional groups can also enhance its ability to adsorb acid gases like H₂S and CO₂ (Plaza et al., 2009).

Enhancing adsorbent performance by combining materials

Binding agents can significantly increase the mechanical strength and efficiency of carbon-based adsorbents. Coal tar pitch, for instance, increases the carbon output by 30–45 % while also improving mechanical resistance (Plaza et al., 2015).

Similarly, introducing resins like phenol-formaldehyde before carbonization enhances the adsorption capacity of the material, making it more effective for biogas purification (Álvarez-Gutiérrez et al., 2014). Additionally, crushed charcoal impregnated with the iron sponge is a structural support and a hydrogen sulfide (H₂S) adsorbent, improving its purification ability (Cherosky & Li, 2013).

In packed-bed adsorption systems, performance can be enhanced by reducing pressure drop through bulking agents. For instance, when using red soil as an adsorbent, incorporating sugarcane mulch—a byproduct of mechanically ground bagasse—improves the adsorption efficiency by modifying the bed structure (Skerman et al., 2017). Figs. 5, 6, and 7 illustrate the preparation, modification, and combination of low-cost materials for biogas purification. Fig. 5 shows the preparation of biomass ash through biomass combustion at 800 °C and the size reduction of red soil to enhance its surface area-to-volume ratio. Fig. 6 details the conversion of coconut shells into activated carbon via carbonization, size reduction, and chemical activation with NaOH or KOH, improving porosity and increasing active sites for better H₂S and

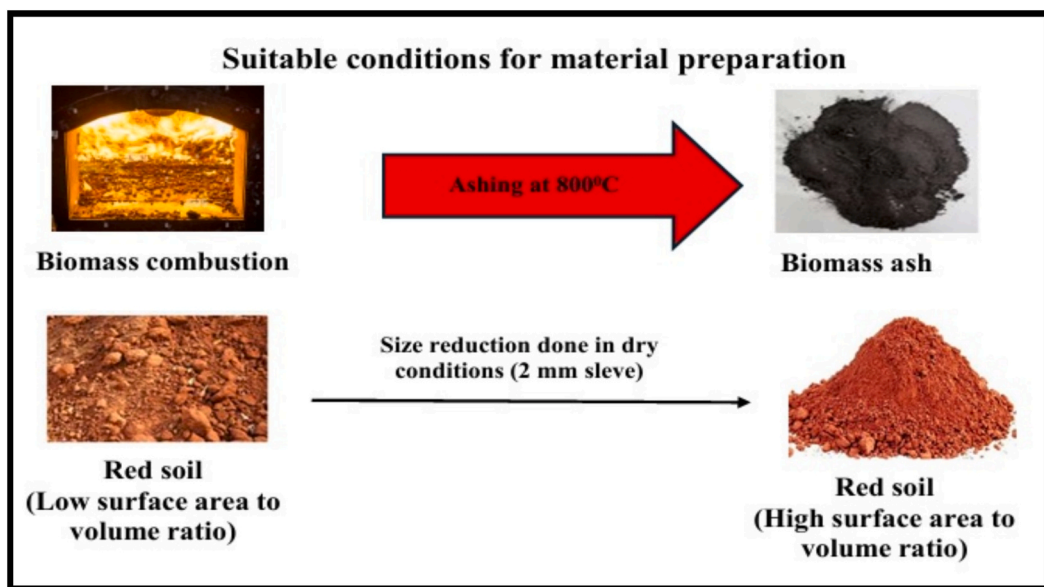


Fig. 5. Schematic representation of optimal conditions for material preparation.

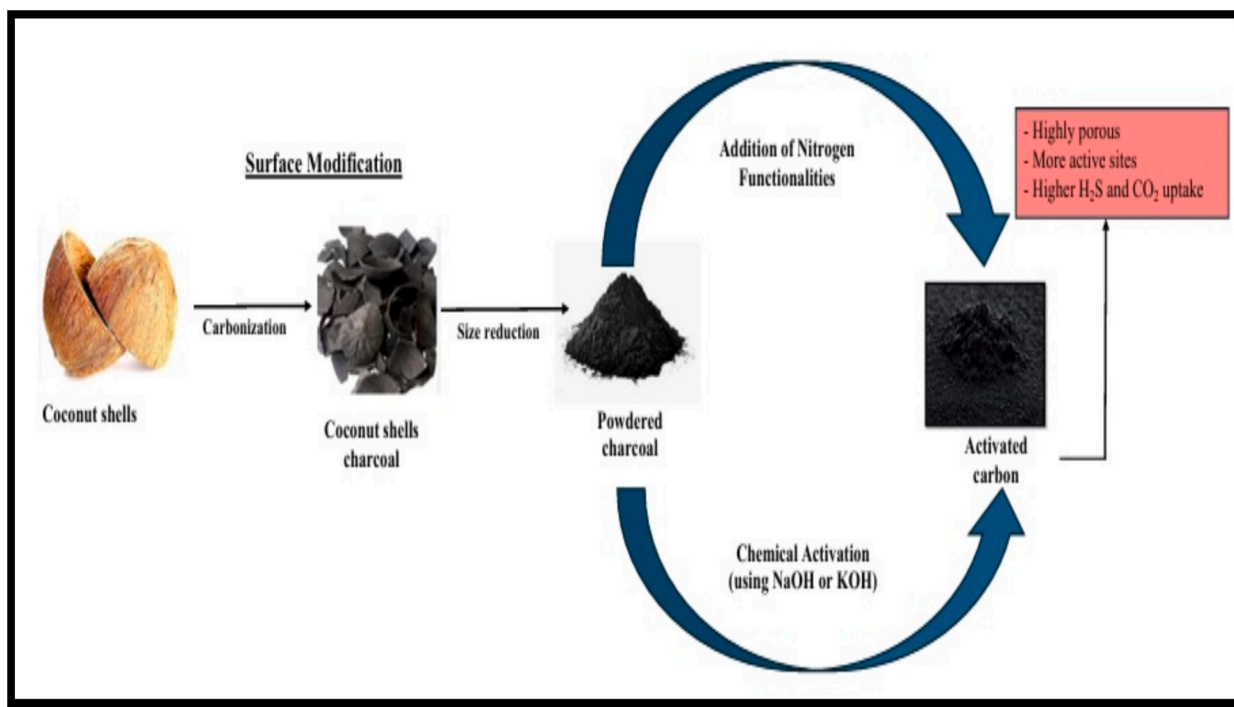


Fig. 6. Depiction of the surface modification of coconut shells to produce activated carbon.

CO₂ uptake. Fig. 7 highlights the combination of absorbent materials, where activated carbon is blended with the iron sponge to enhance CO₂ and H₂S adsorption efficiency. Additionally, sugarcane mulch is integrated into red soil adsorption beds to reduce pressure drop, optimizing the purification system.

Adsorbent regeneration and reuse

It is essential to regenerate and reuse adsorbents to guarantee cost-effectiveness and sustainability in biogas purification. When exposed to air, iron-rich red soils experience a boost in H₂S adsorption capacity, which is attributed to mechanical changes in the particles that enhance

their effectiveness (Skerman et al., 2017). Spent adsorbents that do not contain harmful contaminants can be repurposed for landfilling or even used as fertilizer, reducing environmental waste and supporting agricultural applications (Fernández-Delgado Juárez et al., 2018). Additionally, hybrid adsorbents, including TiO₂/zeolite composites, have shown higher H₂S adsorption efficiency (0.11 mmol/g) than pure zeolite (0.07 mmol/g) or pure TiO₂ (0.09 mmol/g). These materials offer the advantage of low SO₂ emissions while maintaining superior adsorption performance, making them a promising option for biogas purification (Liu et al., 2015).

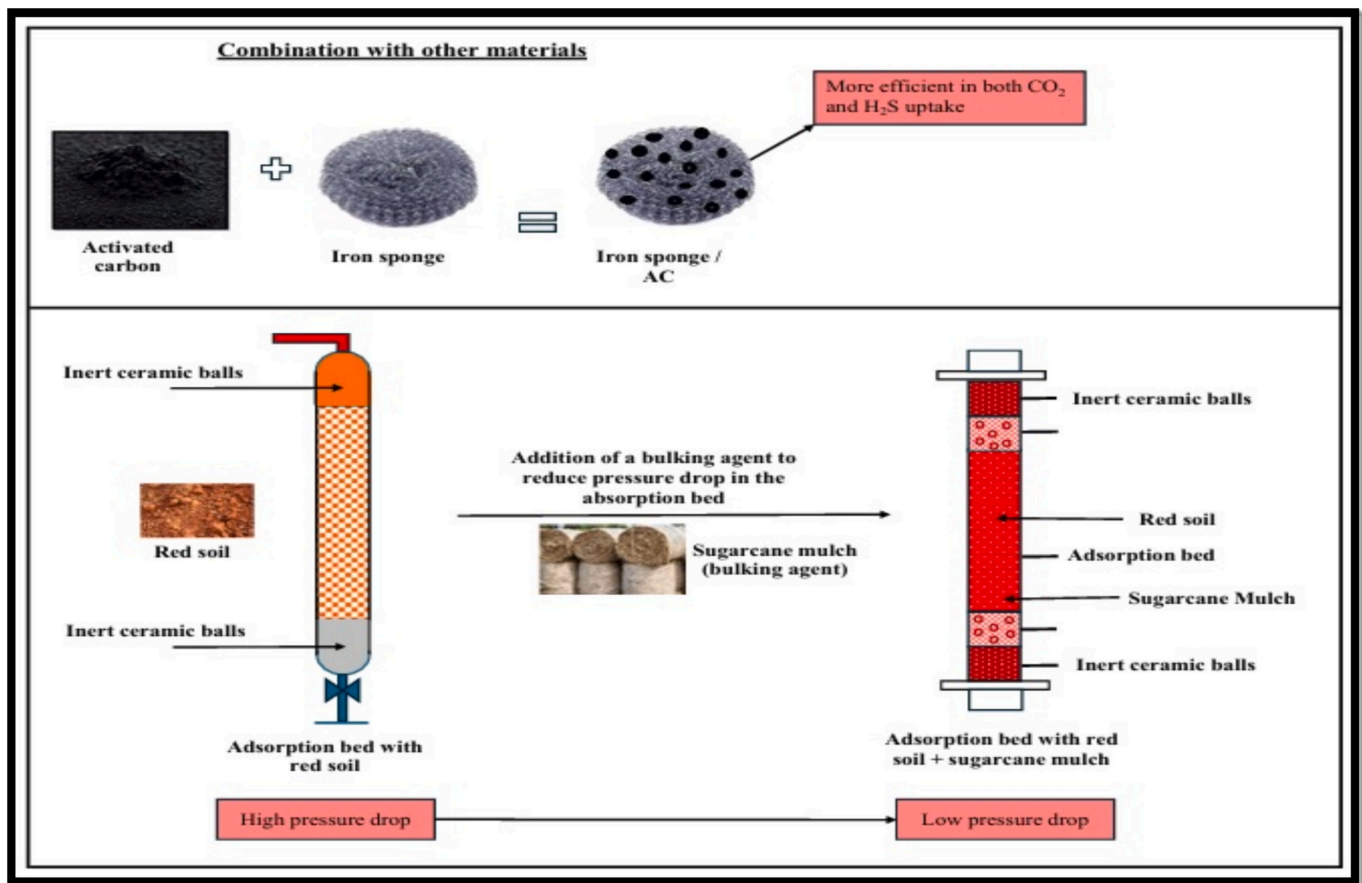


Fig. 7. Schematic representation of fixed-bed adsorption devices.

Conclusion

This study shows that inexpensive materials may effectively remove contaminants from biogas, highlighting their potential for purification. Various adsorbents, such as waste iron chips from workshops, activated carbon from plant waste, iron sponges, biomass ash from incinerators, and iron-rich soils, were assessed. These materials offer a sustainable option for upgrading biogas as they are inexpensive, widely accessible, and likely to avoid major conflicts in most contexts, provided that their collection does not interfere with existing local uses or create new environmental burdens. While comprehensive life cycle assessment (LCA) or supply-chain analyses are limited, preliminary findings suggest their use generally aligns with low-impact and circular economy practices.

Their adoption can support clean energy access, environmental conservation, and efficient waste management. For example, pilot-scale systems using biomass ash and iron-rich adsorbents in rural India and sub-Saharan Africa have displaced 3–5 kWh/day of fossil-derived electricity, avoided approximately 0.7–1.2 tons of CO₂ equivalent emissions annually per household, and valorised up to 500 kg of agricultural or workshop waste. However, broader implementation faces financial hurdles such as upfront costs, lack of micro-financing schemes, and policy gaps around decentralized biogas incentives. The ease of handling, local availability, and straightforward operation of these materials make them promising for widespread use, particularly in developing regions, primarily when favorable environmental and supply-chain conditions exist. However, their performance may vary under specific constraints such as high humidity, cold temperatures, and inconsistent material availability, indicating a need for further sensitivity analysis and context-specific adaptation.

Removing CO₂ and H₂S from raw biogas is crucial for minimizing

health hazards and preventing corrosion in biogas systems. According to research, red soil, waste iron fillings, and cow compost manure have H₂S removal efficiencies of 80–95 %, whereas activated carbon, municipal solid waste, and biomass ash reach CO₂ removal efficiencies of 70–90 %.

To enhance the efficiency of these materials, several improvement strategies are recommended:

- Pre-treating biomass ash to lower the moisture content and optimize adsorption properties.
- Modifying the surface of activated carbon to improve acid gas adsorption.
- Several combined adsorbents, like an iron sponge and activated carbon, can improve the removal effects.
- Regenerating and reusing spent adsorbents to ensure sustainability and cost efficiency.

The study's encouraging results emphasize the necessity of additional investigation and improvement to support large-scale biogas purification applications. The majority of previous research, however, has been carried out on a small scale, highlighting the significance of pilot-scale experiments in producing valuable data for commercial-scale applications. Additionally, further investigations should focus on assessing the techno-economic feasibility of these materials to support their commercial adoption and long-term sustainability in biogas purification.

CRediT authorship contribution statement

Ankur Srivastava: Writing – review & editing. **Pradeep Kumar Meena:** Writing – original draft. **Dinesh Meena:** Data curation. **Sagar Shelare:** Resources. **Chandrika S Wagle:** Visualization.

Consent to participate

Not applicable.

Consent to publish

Not applicable.

Ethical approval

As authors, we would like to tell you that this is our original work, and this paper has not been submitted anywhere except to this journal.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data that support findings of this study are available from corresponding author (Pradeep Kumar Meena) upon reasonable request.

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